

Business Problem Statement

A leading retail company wants to better understand customer shopping behavior to improve sales, customer satisfaction, and long-term loyalty. The company has observed changes in purchasing patterns across demographics, product categories, and sales channels (online vs. offline).

The analysis focuses on identifying how factors such as discounts, customer reviews, seasons, and payment preferences influence purchasing decisions and repeat purchases.

The primary business question is:

“How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?”

Deliverables

- 1. Data Preparation & Modeling (Python):** Clean and transform the raw dataset for analysis.
- 2. Data Analysis (SQL):** Structure the data and analyze customer segments, loyalty, and purchase drivers.
- 3. Visualization & Insights (Power BI):** Build an interactive dashboard highlighting key patterns and trends.
- 4. Report and Presentation:** Summarize key findings and provide actionable business recommendations.
- 5. GitHub Repository:** Maintain a well-structured repository containing the Python scripts, SQL queries, and dashboard files.